Technical Report: Trending YouTube Data Analysis (2025–26)

1. Project Overview

The objective of this project was to build a robust pipeline to process and analyze large-scale YouTube trending data across multiple countries. The analysis covers data cleaning, feature engineering, and extracting insights regarding user engagement and temporal trends.

2. Step-by-Step Technical Analysis

Q1: Data Ingestion and Country Identification

- **Method:** We utilized the `zstandard` library to open compressed `.zst` files as a data stream.
- **Decision:** To save disk space, we read the files directly into memory without physical extraction.
- **Implementation:** During the loop, the first two letters of each filename (e.g., "CA" from `CAvideos`) were extracted to create a new `country` column before concatenating all dataframes into one master table.

Q2: Extracting Tagless Videos

- **Method:** Filtered the dataset for the specific string `[none]`.
- **Decision:** In this dataset, missing tags are explicitly marked by YouTube as `[none]` rather than being empty null values.

Q3: Total Views per Channel

- **Method:** Applied `groupby('channel_title')` followed by the `sum()` aggregate function.
- **Decision:** This allows for a global ranking of content creators by total reach during the 2025–26 period.

Q4: Data Sanitization (Excluded Dataframe)

- **Method:** Created a logical mask to identify rows where comments or ratings were disabled, or the video was removed/errored.
- **Decision:** These videos were moved to an `excluded` dataframe and removed from the main analysis. This ensures that engagement metrics are not skewed by restricted or broken content.

Q5: Feature Engineering: Like Ratio

- **Method:** Used `np.where` to calculate $Like Ratio = \frac{likes}{dislikes}$.
- **Decision:** To prevent "Division by Zero" errors, we handled cases with zero dislikes by assigning `NaN` to ensure code stability.

Q6 & Q7: Temporal Clustering (10-Minute Intervals)

- **Method:** Converted `publish_time` to `datetime` and used `.dt.floor('10min')`.

- **Decision:** By rounding publishing times to the nearest 10-minute block, we transformed granular timestamps into manageable clusters to analyze average likes and dislikes per time slot.

Q8 & Q9: Tag Explosion and Frequency Analysis

- **Method:** Used the `.str.split('|')` and `.explode()` methods.
- **Decision:** Since multiple tags are stored as a single string, we "exploded" them into individual rows. This allowed us to accurately count the frequency of specific keywords like `funny` or `comedy`.

Q10: User Preference Analysis (Tag-Country Pair)

- **Method:** Merged the exploded tag table with the `like_ratio` data.
- **Decision:** This pair-wise analysis identifies which topics resonate most positively with audiences in specific geographic regions.

Q11 & Q13: Identifying Record-Breaking Videos

- **Method:** Utilized `idxmax()` on the `views` column within grouped objects.
- **Decision:** This allowed us to extract the full metadata (title, ID, views) of the "champion" video for every specific day and month per country.

Q12: Date Decomposition

- **Method:** Extracted `year`, `month`, and `day` from the `trending_date`.
- **Decision:** This was necessary to conduct time-series analysis and identify seasonal trends across the years 2025 and 2026.

Q14 & Q15: Category Mapping and "Not Assignable" Identification

- **Method:** Parsed 10 separate JSON files to create a mapping of `category_id` to category names.
- **Decision:** Since category IDs are numeric codes, we mapped them to human-readable names. We also identified videos with IDs not present in a country's JSON file (e.g., 1538 cases in Russia), highlighting metadata discrepancies.

3. Executive Summary

This project successfully implemented a comprehensive pipeline for analyzing global YouTube trends for the 2025–26 period. By moving from raw, compressed data to structured insights, the project achieved:

- **Global Integration:** Merged diverse datasets from 10 countries into a unified analytical framework.
- **Data Integrity:** Improved statistical accuracy by isolating restricted and erroneous videos into an "Excluded" category.
- **Deep Engagement Metrics:** Developed sophisticated metrics like the `Like Ratio` and publishing time clusters to understand audience behavior.
- **Metadata Validation:** Uncovered significant discrepancies in regional category assignments through cross-referencing with JSON metadata.

Conclusion: The infrastructure developed in this project is capable of transforming massive amounts of social media metadata into actionable insights regarding global content trends and user sentiment.